Multi feature classification problem through dimensionality reduction techniques and information theoretic measures for data simplification

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Abstract

We dissected the underlying structure of the dataset by various methods, culminating in a causal model of the features.

1 Data and Preprocessing

We inspected it's properties in the Classical and Rap genres, noticing it contains "?" instances in the tempo column. We converted it's values to numeric and removed missing values's rows.

We notice some durations are invalid numbers, for example negative 1, dropping those entries. The removal of missing values resulted in a 19 per cent shrink in the dataset, which is considerable, and so we applied median inputation on these values.

We remove the obtained-data column, as well as instance-id. The track-name and artist-name could become a feature for language modelling, as we expect some structure to be present distinguishing each genre, but it shouldn't be the main factor. We encode the categorical values like key and mode.

2 Features

Before spliting the dataset with the usual seed 42, we noticed class inbalance was minimal (0.15 per cent relative difference between classical and rap and a maximum of 1.5 per cent between Alternative and Rock) and so determined stratification wouldn't be necessary, as we don't expect a bias to develop inside the classifier.

Initially we look closer at the distribution of numeric features, namely instrumentalness, that between these two classes will probably be the least informative, as it is squashed to zero on all datapoints, as seen in Figure 1. We point out that features like energy, danceability and in particular acousticness clearly differentiate the two genres, conjecturing that they are designed to capture musical caractheristics that may correlate strongly with genre. That being the case, a simple decision tree algorithm may already capture a meaningful amount of the separation, although ensembles are likely to perform better, as the cost of the latter scales with the number of genres due to the necessity for informed heuristics. Adding to that, features like duration and liveness are very concentrated throughout the different genres, making them less useful for these kinds of approaches. We could construct a composite score reflecting the degree to which a track exhibits characteristics asso-

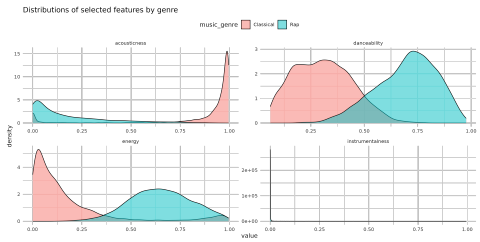

Figure 1: Distributions of acousticness, danceability, energy and instrumentalness over the Rap and Classical genres

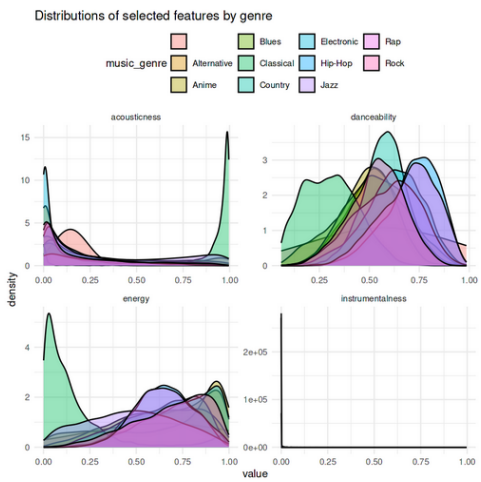

Figure 2: Distributions of acousticness, danceability, energy and instrumentalness over all genres

ciated with rap, that could become a rapiness feature, though in practice a supervised model shoul learn such interactions more reliably.

We notice that some features are roughly unimodal and bell-shaped, namely danceability and popularity, whereas many others divide the data into two clusters of very high values and very low values, like acoustisness, taking discrete binary values essentially, being highly skewed. Besides these some are bounded, or concentrated near zero, like duration and instrumentalness, respectively.

3 Methods and Results

3.1 Information theoretic aware Feature Selection

Eight sequential forward selection methods based on mutual information were implemented and applied to the 11 scaled numeric training features: MIM, MIFS, mRMR, maxMIFS, CIFE (implemented here as its normalised variant DISR), CMIM, JMI, and DMIM [2, 3, 4]. All methods share the same SFS skeleton: at each step the candidate feature X_i maximising a method-specific scoring function $J(X_i)$ is appended to the selected set S. The functions differ in how they trade off relevance $I(X_i; C)$ against redundancy with already-selected features, and whether they incorporate class-conditional complementarity $I(X_i; X_s \mid C)$.

MI was estimated via equal-frequency histogram discretisation with $k = \lceil \sqrt{N} \rceil$ bins [4] for the univariate case, with analogous bin counts for joint and conditional distributions.

The number of features to retain (k_{final}) was determined by inspecting figure 3.

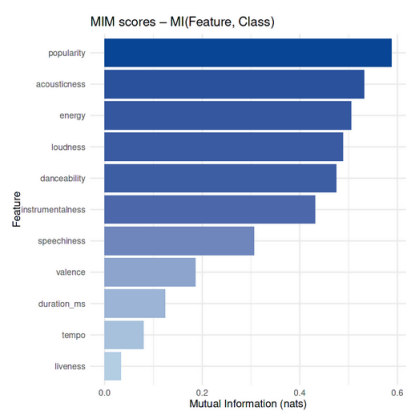

Figure 3: MIM relevance scores for all 11 numeric features. The elbow after position 6 justifies retaining $k_{\text{final}} = 6$ features across all methods.

Table 1 reports the full feature rankings produced by all eight methods. There is broad

agreement on the top positions: `popularity` and `loudness` are consistently placed in the top two or three, reflecting the strong genre-proxy role of production-era loudness normalisation and playlist-driven popularity. `Instrumentalness`, `danceability`, and `speechiness` appear in the top six for most methods, capturing the known acoustic signatures of Classical (near-binary high instrumentalness), Rap and Hip-Hop (high speechiness and danceability), and Electronic music. The key methodological differences emerge in the middle and lower ranks, discussed in Section 6.

Table 1: Feature rankings for five representative methods. maxMIFS, DISR, and JMI are omitted as their rankings are subsumed by the methods shown

Rk	MIM	MIFS	mRMR	CMIM	DMIM
1	pop.	pop.	pop.	pop.	pop.
2	loud.	dance.	loud.	loud.	loud.
3	acous.	loud.	instr.	instr.	instr.
4	instr.	instr.	dance.	dance.	dance.
5	energ.	acous.	speech.	speech.	speech.
6	speech.	speech.	energ.	val.	acous.
7	val.	val.	val.	acous.	val.
8	dance.	energ.	acous.	energ.	energ.
9	dur.	dur.	dur.	dur.	dur.
10	tempo	tempo	tempo	tempo	tempo
11	live.	live.	live.	live.	live.

Figure 4 makes the between-method divergences explicit by showing, for each feature, which methods include it in their top-6 subset.

The bottom three features (duration_ms, tempo, liveness) are never selected, confirming their low marginal relevance. The key divergences are in positions 6–8: MIFS uniquely promotes `danceability` to rank 2 because its unscaled penalty pushes the method away from `loudness`-correlated features early in the selection process; CMIM uniquely selects `valence`, the only method to do so, reflecting its complementarity-aware scoring.

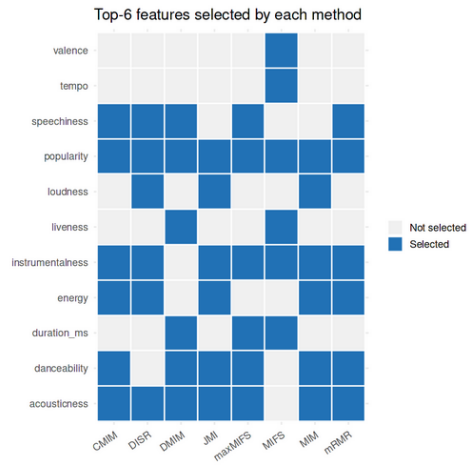

Figure 4: Top 6 features selected in classification of all genres across all information theoretic measures

3.2 PCA

Explain PCA.

$$X = UDV^\top \tag{1}$$

$$\sum_{i,j=1}^{N} a_{ij}\partial_i\phi\partial_j\phi = 0, \tag{2}$$

$$K(\mathbf{x}, \mathbf{x}') = \tanh(k\mathbf{x}^\top\mathbf{x}' + c) \tag{3}$$

$$K(\mathbf{x}, \mathbf{x}') = \phi(\mathbf{x})^\top\phi(\mathbf{x}') = \langle\phi(\mathbf{x}), \phi(\mathbf{x}')\rangle \tag{4}$$

3.3 Kernel PCA

Explain kernel PCA.

4 Classification: Random Forest

4.1 Training and Evaluation Protocol

A Random Forest (RF) classifier was trained with 500 trees, $m = \lfloor\sqrt{p}\rfloor$ features sampled

at each split, and no constraint on minimum leaf size (full-depth trees to minimise bias). Training was performed separately for each dimensionality-reduction strategy: (i) the full original feature set (11 numeric + 2 categorical); (ii) the PCA-reduced representation; (iii) the Kernel PCA-reduced representation and (iv) the subset selected by each of the eight MI methods. All models were evaluated on the held-out test set ($N_{\text{test}} = 15\,000$) using Accuracy, Macro Recall, Macro Precision, and Macro F1, where macro-averaging weights each of the 10 classes equally.

4.2 Results

Table 2 consolidates all strategies and metrics. The full-feature RF leads on every metric. All eight MI FS methods form a tight cluster between Macro F1 0.461 and 0.523, substantially above PCA and Kernel PCA (0.387 and 0.384).

Table 2: RF performance across all strategies. FS methods retain 83%–94% of the full-model Macro F1 with 6 features.

Method / config	Acc.	Mac. Re	Mac. Pr	Mac. F1
all features	0.5566	0.5569	0.5593	**0.5558**
CMIM	0.5229	0.5231	0.5258	0.5225
DISR	0.5111	0.5111	0.5138	0.5105
maxMIFS	0.5089	0.5090	0.5110	0.5081
mRMR	0.5084	0.5085	0.5108	0.5072
MIM	0.4909	0.4908	0.4951	0.4912
JMI	0.4883	0.4883	0.4928	0.4888
DMIM	0.4881	0.4881	0.4907	0.4867
MIFS	0.4625	0.4626	0.4644	0.4608
PCA	0.3904	0.3898	0.3872	0.3871
Kernel PCA	0.3868	0.3861	0.3845	0.3837

Original dataset. Macro F1 **0.556**; OOB error 0.451 is consistent with the test error, confirming no substantial overfitting. The confusion matrix (Figure **??**) reveals characteristic error clusters: Hip-Hop and Rap are frequently confused (overlapping speechiness and danceability); Blues and Jazz share a cluster (common harmonic texture); Classical is the most reliably identified genre due to its near-binary instrumentalness signature.

Feature selection methods. CMIM achieves the best FS performance (Macro F1 0.5225, retaining 93.9% of the full-model score with only 6 features). Its scale-invariant penalisation via maximisation prevents the relevance signal from being overwhelmed. MIFS is the weakest (0.4608), directly validating [4]: its unscaled penalty $\sum_s I(X_i; X_s)$ grows linearly with $|S|$ and eventually dominates, selecting maximally independent but less genre-informative features. mRMR and maxMIFS produce near-identical results (0.5072 and 0.5081), consistent with their algorithmic similarity. JMI and DMIM underperform their theoretical standing, likely because conditional MI estimates $I(X_i; X_s \mid C)$ are noisier on fine probability tables required by 10-class discretisation [3]. Confusion matrices for the best and worst FS methods are shown in Figure **??**.

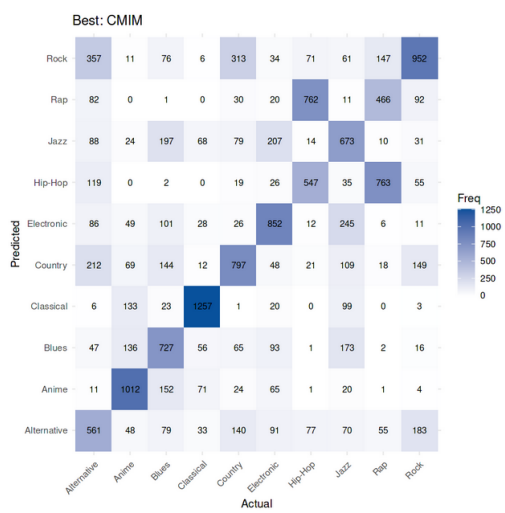

Figure 5: Confusion matrix for CMIM (Macro F1 = 0.522). Hip-Hop/Rap and Blues/Jazz remain the dominant confusion pairs.

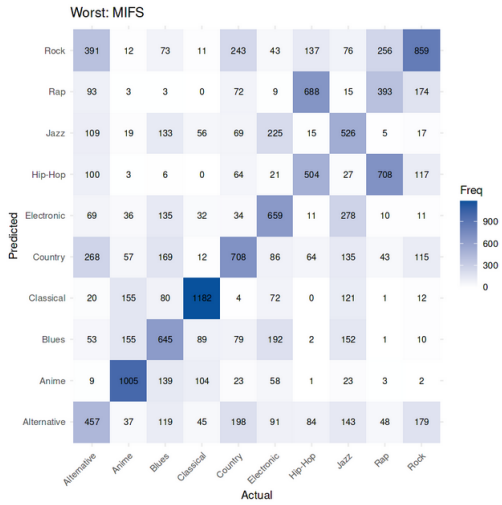

Figure 6: Confusion matrix for MIFS (Macro F1 = 0.461). Increased Hip-Hop/Rap confusion reflects a less discriminative feature subset.

Dimensionality reduction. PCA and Kernel PCA substantially underperform all MI-based approaches.

4.3 Variable Importance vs. MI Rankings

Table 3 compares RF permutation importance rank with MIM (univariate MI) rank. The Spearman rank correlation is 0.691, indicating moderate but imperfect agreement.

The two most instructive discrepancies are `loudness` (MIM rank 2, RF rank 5) and `danceability` (MIM rank 6, RF rank 3). Loudness carries high marginal MI but is largely redundant once `popularity` and `energy` are in the ensemble—a relationship MIM cannot detect. Danceability becomes more informative in the context of other features than its marginal MI implies, a complementarity effect that CMIM and DMIM are specifically designed to capture.

Table 3: RF permutation importance rank vs. MIM rank. Spearman $\rho = 0.691$. Discrepancies ≥ 3 positions are italicised.

Feature	RF Rank	MIM Rank
popularity	1	1
instrumentalness	2	4
danceability	3	*6*
speechiness	4	*7*
loudness	5	*2*
valence	6	*8*
acousticness	7	3
duration_ms	8	9
energy	9	5
tempo	11	10
liveness	12	11

5 Causal Discovery and Interventional Analysis

5.1 Motivation

Standard classification evaluation does not answer wether the learned feature–label associations are causally stable or spurious correlations that break under distribution shift. Motivated by [5], we complement the benchmarks with three causal tools: a data-driven causal skeleton (PC algorithm), interventional response curves via Pearl's do-calculus, and an Interventional Robustness Score (IRS) proxy.

5.2 Causal Graph via the PC Algorithm

The PC algorithm was applied to the 11 scaled numeric features plus an integer-encoded genre node, using the Gaussian CI test ($\alpha = 0.005$). The output is a CPDAG over 12 nodes.

The graph recovers several acoustically interpretable relationships. Energy and loudness are connected by an oriented edge (physical amplitude–loudness relationship). Instrumentalness connects to multiple genre dimensions (near-binary separator for Classical vs. vocal

genres). Speechiness and danceability share an undirected edge, consistent with both being elevated in Hip-Hop and Rap for production-style reasons rather than a direct causal chain. Several edges remain unoriented—the correct epistemic statement for purely observational data.

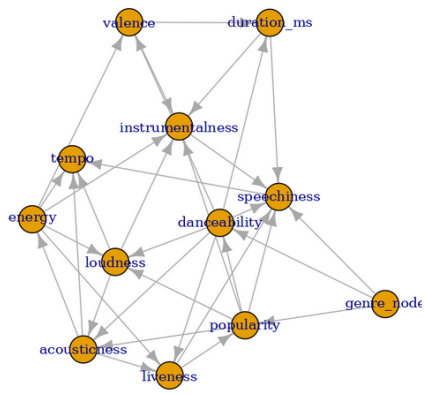

Figure 7: PC-algorithm CPDAG over 11 audio features and the integer-encoded `genre_node` ($\alpha = 0.005$, Gaussian CI test). Oriented edges indicate an identifiable causal direction; unoriented edges are non-identifiable from observational data alone.

5.3 Interventional Response Curves

For three feature–genre pairs we computed the observational curve $P(\text{Genre} \mid X = t)$ and the interventional curve $P(\text{Genre} \mid \mathbf{do}(X = t))$ as t is swept from the 5th to the 95th percentile. The observational curve is estimated by conditioning on the nearest training point to t; the interventional curve uses the backdoor adjustment:

$$P(Y \mid \mathbf{do}(X = t)) = \frac{1}{N} \sum_{i=1}^{N} \hat{P}(Y \mid X = t, \mathbf{Z} = \mathbf{z}_i), \tag{5}$$

marginalising over the empirical distribution of remaining features $\mathbf{Z}$, as justified by Proposition 1(f) of [5]. The RF serves as $\hat{P}$. The single-observation conditioning makes the observational curves jagged; the interventional curves are smooth by construction, and this contrast itself reveals the confounding structure.

Experiment A — Instrumentalness → Classical. Both curves rise with instrumentalness, but the observational curve is markedly more erratic, indicating confounding from co-occurring Classical markers (low loudness, low tempo variability). The backdoor-adjusted curve isolates the direct causal effect.

Experiment B — Speechiness → Rap. The interventional curve (Figure 9) shows a smooth monotone increase in predicted Rap probability, confirming a genuine causal contribution after confounders are removed. The smaller gap between the two curves relative to Experiment A suggests speechiness is a more isolated driver of Rap, with fewer co-occurring confounders.

Experiment C — Popularity → Classical. The interventional curve is nearly flat (≈ 0.10–0.15 across the full range), the key diagnostic result: popularity does not causally determine genre. The observational correlation exists because Classical music has characteristically low streaming popularity, but forcing a Classical track to high popularity would not change its audio content and therefore not change the classifier's prediction.

5.4 Interventional Robustness Scores

For method m, the IRS proxy is:

$$\widehat{\text{IRS}}_m = 1 - \frac{\overline{\text{PIDA}}_m}{\text{sd}[\hat{P}(\text{Rap} \mid X)]}, \tag{6}$$

where $\overline{\text{PIDA}}_m$ is the mean, over nuisance features in the selected subset, of the maximum absolute shift in predicted Rap probability when that feature is swept through a 5-point quantile grid while the top-ranked feature is held at its median.

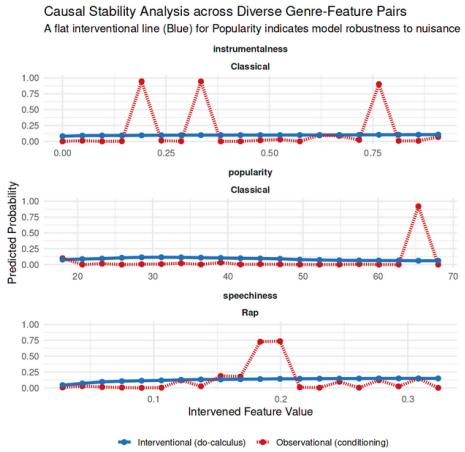

Figure 8: Observational vs. interventional probability curves for three feature–genre pairs. Jagged red curves reflect single-point conditioning; smooth blue curves are the backdoor-adjusted causal estimates (Eq. 5).

Table 4 reports IRS and Macro F1 for all eight methods, revealing a clear and systematic *inverse* relationship between the two quantities.

MIFS achieves the highest IRS (0.978) because its independence-maximising selection leaves no single feature dominant, so nuisance interventions cause only small prediction shifts. The cost is accuracy (Macro F1 0.461): independent features carry less concentrated genre signal. CMIM achieves the lowest IRS (0.964) because complementary feature pairs are more causally intertwined through shared genre determinants, making the classifier more sensitive to single-feature interventions—which is *not* a failure, since a well-calibrated classifier should respond to interventions on causally active features.

The practical implication is a two-step workflow: use CMIM or DISR to maximise classification accuracy, then audit the selected subset with the IRS. Features with high IRS but low permutation importance are pruning candidates; features with low IRS but high importance are causally active and should be retained.

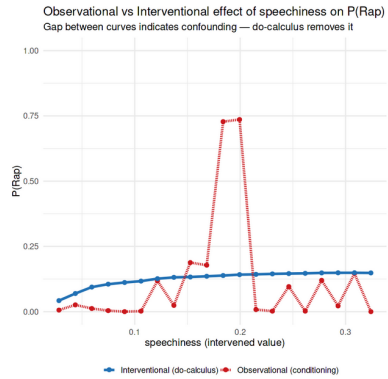

Figure 9: Detailed view: speechiness → Rap. The monotone rise of the interventional curve (blue) confirms speechiness as a causally active feature for the Rap genre.

6 Discussion and Conclusion

6.1 Feature Selection vs. Full-Feature Baseline

The full-feature RF (Macro F1 0.556) outperforms all FS models. This is expected when the feature count is modest and the training set is large (35 000 observations): the RF is not harmed by weak features. The value of FS here lies in interpretability, computational efficiency, and causal auditing.

6.2 The Relevance–Redundancy Balance

The empirical ranking (CMIM > DISR > maxMIFS ≈ mRMR > MIM > JMI ≈ DMIM > MIFS) closely mirrors the theoretical predictions of [2], with one notable exception: DMIM underperforms JMI. The authors of [3] attribute occasional reversals to MI estimation noise; with 10 classes and continuous features requiring discretisation, conditional MI estimates $I(X_i; X_s \mid C)$ rest on finer probability tables and are therefore noisier, dampening the DMIM advantage.

The single most strongly theory-predicted result—MIFS failure—is reproduced here on a

Table 4: IRS proxy and Macro F1 by FS method (sorted by IRS, descending). All methods exceed IRS 0.96. The inverse relationship with accuracy is a structural consequence of each method's redundancy-control philosophy.

Method	IRS proxy	Macro F1
MIFS	0.978	0.461
DMIM	0.971	0.487
MIM	0.969	0.491
JMI	0.969	0.489
mRMR	0.965	0.507
maxMIFS	0.965	0.508
DISR	0.964	0.511
CMIM	0.964	0.523

real 10-class dataset, providing an empirical counterpart to the theoretical proof in [4] that MIFS with $\beta = 1$ does not balance the relevance–redundancy trade-off as $|S|$ grows.

6.3 Causal Perspective

The PC algorithm identifies a plausible causal skeleton with acoustically interpretable edges. The do-calculus experiments distinguish causally active features (instrumentalness, speechiness) from a nuisance feature (popularity)—a distinction invisible from classification accuracy alone. The IRS analysis shows that the accuracy–robustness trade-off between CMIM and MIFS is a structural consequence of their selection philosophies: complementarity-seeking methods select causally intertwined features (lower IRS, higher accuracy); independence-seeking methods select causally isolated features (higher IRS, lower accuracy).

References

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